Data Analyst – Securitisation

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Power BI Risk Assessment & Portfolio Analytics Dashboard

Project 1A | Supporting Submission Report

This report documents the completed Power BI dashboard and the analytical measures and visuals developed for the securitisation dataset.

1. Project Overview

The project focuses on building a securitisation risk assessment dashboard using assumed bank/ERP-style auto-loan data. The analysis covers portfolio exposure, delinquency, write-offs, servicer performance, IFRS 9 stage distribution, monthly losses and vintage performance.

Primary tool: Microsoft Power BI

Focus: Credit-risk monitoring, portfolio analytics and securitisation performance reporting.

2. Data Sources

Dataset	Purpose
auto_loan_securitisation_data	Loan-level portfolio, balance, servicer and IFRS 9 information
dpd_snapshot_history	Monthly DPD, delinquency and loan-status history
dynamic_loss_monthly	Monthly pool loss and performance measures
static_pool_vintage_data	Vintage/cohort performance and cumulative loss analysis

3. Dashboard KPIs

KPI	Value
Total Loan Balance	317.79M
Total Loans	500
Average Loan Balance	635.57K
Delinquent Loans	275
Delinquency Rate	55.00%
Written Off Loans	6
Write-Off Rate	1.20%

4. Dashboard Visuals

DPD Bucket Distribution: Shows observed records across delinquency buckets. Since the DPD table contains monthly snapshots, historical bucket counts can overlap and should not be interpreted as one-period totals.

Monthly Net Loss Trend: Tracks monthly net losses across the reporting period.

Loan Balance by Servicer: Compares current portfolio balance handled by each servicer.

Delinquent Loans by Servicer: Compares delinquent loan counts across servicers.

IFRS 9 Stage Distribution: Shows the loan distribution across IFRS 9 credit-risk stages.

Vintage Cumulative Loss Rate: Tracks cumulative loss rate by months on book.

Monthly Net Loss Rate: Shows monthly loss-rate movement from the dynamic pool-performance data.

30+ DPD by Months on Book: Tracks current 30+ DPD performance as the portfolio seasons.

5. DAX Measures Implemented

Measure	Purpose
Total Loan Balance	Sum of current loan balances
Total Loans	Distinct count of LoanID
Average Loan Balance	Average current loan balance
Delinquent Loans	Distinct loans with DPD_Days > 0
Delinquency Rate	Delinquent Loans ÷ Total Loans
Written Off Loans	Distinct loans with WriteOffFlag = TRUE
Write-Off Rate	Written Off Loans ÷ Total Loans

6. Interactive Analysis

Two slicers were added: **Servicer** and **PoolID**. These allow the dashboard user to filter the analysis by servicing entity or securitisation pool.

The model includes a relationship between DPD snapshot history and the loan-level portfolio table, enabling delinquency measures to be analysed in the loan context.

7. Risk & Business Interpretation

• Portfolio balance and average balance provide an exposure view of the securitised pool.

• Delinquency metrics help identify payment-performance issues.

• Write-off metrics show severe loss-status outcomes present in the available history.

• Servicer comparisons support investigation of exposure and delinquency concentration.

• IFRS 9 stage distribution provides a credit-risk classification view.

• Monthly loss and vintage analysis help monitor both short-term movement and portfolio seasoning.

8. Conclusion

The completed Power BI dashboard brings together loan-level, delinquency, monthly loss and vintage datasets into a single securitisation monitoring view. It provides interactive KPIs, portfolio segmentation, credit-risk indicators and trend analysis, forming a foundation for deeper stress testing, ECL modelling and investor reporting.

9.Link

Zetheta_Data_Analyst_Securitisation.pbix

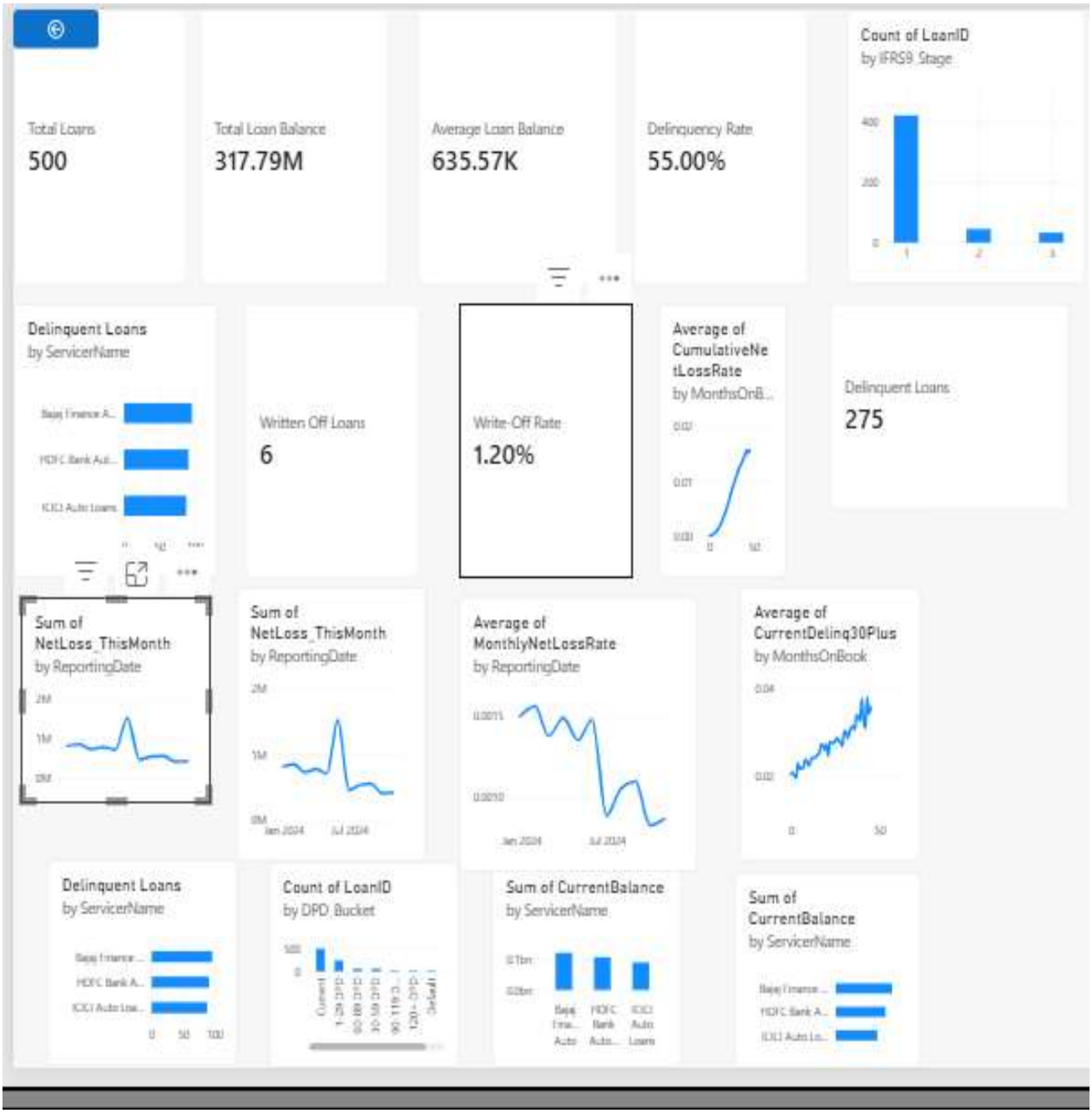


Figure 1: Power BI Securitisation Risk Assessment Dashboard showing portfolio KPIs, delinquency analysis, loss trends, servicer performance, IFRS 9 stage distribution, and vintage performance.